

University of Nottingham Malaysia

SCHOOL OF COMPUTER SCIENCE
A LEVEL 1 MODULE, AUTUMN SEMESTER 2021-2022

COMPUTER FUNDAMENTALS (COMP1027)

Time allowed: **ONE** Hour

An additional 30 minutes to cover IT issues and upload time

Answer ALL questions

Submit your answers containing all the work you wish to have marked as a **single PDF file, with each page in the correct orientation, to the appropriate Dropbox on the module's Moodle page.**

Use the standard naming convention for your document: [Student ID]_[Module Code]. Write your student ID number at the top of each page of your answers. Do not include your name.

Although you may use any notes or resources you wish to help you complete this open-book examination, the academic misconduct policy that applies to your coursework also applies here. You must be careful to avoid plagiarism, collusion or false authorship. This statement refers to, and does not replace, the University policy which stipulates severe penalties for academic misconduct.

Staff are not permitted to answer assessment or teaching queries during the period in which your open-book examination is live. If you spot what you think may be an error on the exam paper, note this in your submission but answer the question as written.

IMPORTANT: Lateness penalties will not be applied, and late submission will not be accepted.

Marks available for sections of questions are shown in brackets in the right-hand margin

No calculators are permitted in this examination

1. Number Representation and Binary Arithmetic [18%]

- a) Given Table 1, calculate the 6 values labelled *a-f*. Your representations for Binary, 1's Complement and 2's Complement must be in 12-bits width. (6)

Table 1.

Decimal	Binary	1's Complement	2's Complement
1010	<i>a</i>	<i>b</i>	<i>c</i>
<i>f</i>	<i>e</i>	<i>d</i>	1011 1011 1010

- b) Given the following values of *x*, *y* and *z* below (in Decimal numbers):

- $x = -1027$,
- $y = 540$,
- $z = -43$

Calculate, showing your working in details, the binary representation in 12-bit *TWO's* complement. (6)

- c) Using your **12-bits** *TWO's* complement answers in (b), calculate the following the sums. Show your working in detail. Your answer should be in 12-bits width. (6)
- i. $x + y$
 - ii. $y - z$
 - iii. $x + z$

2. Boolean Algebra & Digital Logic [30%]

- a) In the lectures, you have been introduced to **three** basic & **two** universal gates. List them by drawing their respective logic diagram and truth table with A, B as inputs and Y as the output. Please label the inputs and output in your logic diagram. (15)

- b) Given the below **truth table** in Table 2:

Table 2.

X	Y	Z	F
1	1	1	1
1	1	0	0
1	0	1	1
1	0	0	0
0	1	1	1
0	1	0	0
0	0	1	0
0	0	0	0

- i. Write out the expression (pre-simplified) (3)

Identity Name	AND Form	OR Form
Identity Law	$1x = x$	$0+x = x$
Null (or Dominance) Law	$0x = 0$	$1+x = 1$
Idempotent Law	$xx = x$	$x+x = x$
Inverse Law	$x\bar{x} = 0$	$x+\bar{x} = 1$
Commutative Law	$xy = yx$	$x+y = y+x$
Associative Law	$(xy)z = x(yz)$	$(x+y)+z = x+(y+z)$
Distributive Law	$x+yz = (x+y)(x+z)$	$x(y+z) = xy+xz$
Absorption Law	$x(x+y) = x$	$x+xy = x$
DeMorgan's Law	$(\bar{x}\bar{y}) = \overline{x+y}$	$(\bar{x}+\bar{y}) = \overline{xy}$
Double Complement Law	$\overline{\bar{x}} = x$	

Extracted from: L. Null, L. and J. Lobur, *The essentials of computer organisation and architecture*, Jones & Bartlett Publishers, 2003 (Fig. 1.3, P.26)

Figure 1.

- ii. Using the given Basic Identities of Boolean Algebra in Figure 1, simplify your expression in b(i). Detail each step. (3)
- iii. Based on your answer in (b), draw the Logic Circuit Diagram to implement the circuit (using the relevant combinations of ONLY basic gates). (6)

3. Adders, MUX and ALU [27%]

- a. Explain the **significant** difference between:
 - i. Combinatorial Logic and Sequential Logic (3)
 - ii. Multiplexer (MUX) and De-multiplexer (DEMUX) (3)
- b. List down the differences that sets a Half Adder and Full Adder apart in terms of functionality and gates utilised. (6)
- c. For the ALU design (given in Figure 2), show (with *full working details*) how this ALU can achieve each of the listed output functionalities (*out*) according to the given corresponding control values in Table 3 (on page 4).

In your working details, you need to state what each of the variables (x_1 , x_2 , y_1 , y_2 , K_1 , K_2 , Out_1 , Out) represents/contains in each case (a, b, c; in the table below)

Your answers should be in 4-bits width: (15)

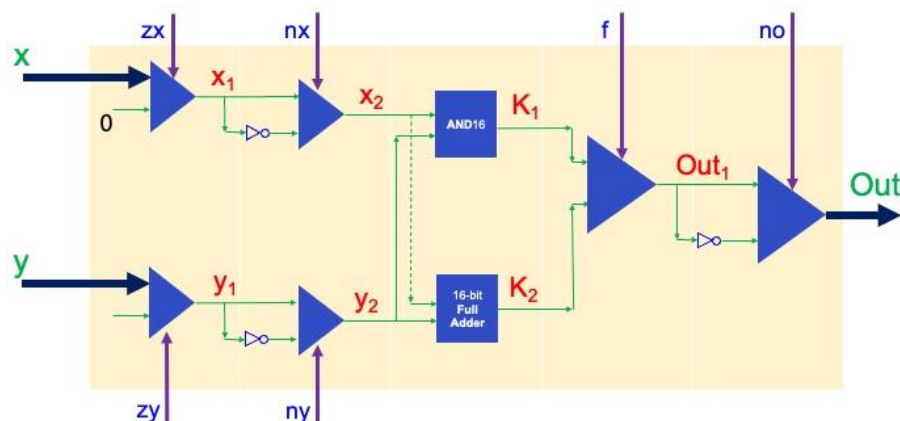


Figure 2.

P.T.O

Table 3.

	zx	nx	Zy	ny	f	no	Out
a.	0	0	1	1	0	1	!x
b.	1	1	0	1	1	1	y+1
c.	0	0	0	1	1	1	y - x

Hint: $-A = A\sim + 1$ { where $A\sim$ is the bit-wise Not(A) }.

4. CPU Instructions and Networks [25%]

- a. Based on your understanding of the Hack Computer, describe the **Fetch-Execute Cycle** process for A- or C- Instructions. (4)

Table 4.

	Layer Name	Provide/Facilitate
7	-	-
6	-	-
5	-	-
4	-	-
3	-	-
2	-	-
1	-	-

- b. Table 4 above shows the OSI Network stack:
- Give the names for layers 1 till 7 and list what each layer provide/facilitate. (14)
 - Give **one** example of common protocols for layers two, three, four and seven. (4)
- c. Describe the **main difference** between and IPv4 address and an IPv6 address and why it is necessary to move from IPv4 to IPv6. (3)

END